Multilayer Perceptron

Dataset: MNIST
Each piece of data is a 28*28(784 features) grayscale image of a handwritten digit.
Training-set includes 60000 images; Test-set includes 10000 images

Number of input layer neuron: 784 (size: 28*28)
Number of output layer neuron: 10 (class: 10)

Data processing

In [10]:
```
import pandas as pd
import torch
```

In [11]:
```
# complete the code here:
# Read .csv files to load data
train_path = './mnist_train.csv'
test_path = './mnist_test.csv'
train_data = pd.read_csv(train_path,header=None)
test_data = pd.read_csv(test_path,header=None)
```

In [12]:
```
# complete the code here:
# 1. print the shape of the train and test data
# 2. explain the meaning of shapes in the comments
print("Train data shape:", train_data.shape)
print("Test data shape:", test_data.shape)
#(60000,785) means the train data has 60000 samples, and each sample has 785 dim
#(10000,785) means the train data has 10000 samples, and each sample has 785 dim
```

```
Train data shape: (60000, 785)
Test data shape: (10000, 785)
```

In [13]:
```
y_train = train_data.iloc[:, 0].values  # get labels
x_train = train_data.iloc[:, 1:].values # get features

y_test = test_data.iloc[:, 0].values # get labels
x_test = test_data.iloc[:, 1:].values # get features
```

In [14]:
```
# convert data to tensor
x_train = torch.tensor(x_train, dtype=torch.float32)
y_train = torch.tensor(y_train, dtype=torch.long)

x_test = torch.tensor(x_test, dtype=torch.float32)
y_test = torch.tensor(y_test, dtype=torch.long)
```

In [15]:
```
print(x_train.shape)
print(y_train.shape)
print(x_test.shape)
print(y_test.shape)
```

```
torch.Size([60000, 784])
torch.Size([60000])
torch.Size([10000, 784])
torch.Size([10000])
```

Building Neural Networks

In [17]:
```python
import torch.nn as nn
import torch.nn.functional as F
from tqdm.auto import tqdm
```

In [18]:
```python
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
```

In [19]:
```python
# explain the details of the neural network designed here by adding comments

class MNIST(nn.Module):

    def __init__(self):
        super().__init__()
        self.fc1 = nn.Linear(28*28, 1000) #define fc1, transform 28*28 input fea
        self.fc2 = nn.Linear(1000, 500) #define fc2, transform 1000 input featur
        self.fc3 = nn.Linear(500, 10) #define fc3, transform 500 input features

    def forward(self, x):
        x = F.relu(self.fc1(x)) #apply fc2 with ReLU
        x = F.relu(self.fc2(x)) #apply fc2 with ReLU
        x = self.fc3(x) #get logits from fc3
        return x
```

In [20]:
```python
mnist = MNIST().to(device) # instantiation
print(mnist)
```

```
MNIST(
  (fc1): Linear(in_features=784, out_features=1000, bias=True)
  (fc2): Linear(in_features=1000, out_features=500, bias=True)
  (fc3): Linear(in_features=500, out_features=10, bias=True)
)
```

In [21]:
```python
# this code blcok is the training process
# define learning rate
lr = 0.001

losses = []

for epoch in range(100):
    x_train = x_train.to(device)
    y_train = y_train.to(device)
    y_pred = mnist(x_train)

    # calculate the loss
    loss = F.cross_entropy(y_pred, y_train)

    # backpropagation
    loss.backward()

    # Updating the gradient
    with torch.no_grad():
        for p in mnist.parameters():
            p -= lr * p.grad
```

```
        mnist.zero_grad() # Avoid gradient accumulation

        print(f'Epoch {epoch}, Loss: {loss.item()}')
        losses.append(loss.item())
```

```
Epoch 0, Loss: 15.303167343139648
Epoch 1, Loss: 17.549877166748047
Epoch 2, Loss: 16.211673736572266
Epoch 3, Loss: 12.923460006713867
Epoch 4, Loss: 8.866988182067871
Epoch 5, Loss: 5.175592422485352
Epoch 6, Loss: 4.0699310302734375
Epoch 7, Loss: 2.75365948677063
Epoch 8, Loss: 1.5463194847106934
Epoch 9, Loss: 1.2549186944961548
Epoch 10, Loss: 1.1123673915863037
Epoch 11, Loss: 1.0464227199554443
Epoch 12, Loss: 0.9830663800239563
Epoch 13, Loss: 0.9428256750106812
Epoch 14, Loss: 0.903769314289093
Epoch 15, Loss: 0.8735325336456299
Epoch 16, Loss: 0.8458825349807739
Epoch 17, Loss: 0.8220047950744629
Epoch 18, Loss: 0.800478458404541
Epoch 19, Loss: 0.7811090350151062
Epoch 20, Loss: 0.7634255886077881
Epoch 21, Loss: 0.7471451163291931
Epoch 22, Loss: 0.7320547699928284
Epoch 23, Loss: 0.7179749608039856
Epoch 24, Loss: 0.7047577500343323
Epoch 25, Loss: 0.692325234413147
Epoch 26, Loss: 0.6805911064147949
Epoch 27, Loss: 0.6694783568382263
Epoch 28, Loss: 0.6589412689208984
Epoch 29, Loss: 0.6489337086677551
Epoch 30, Loss: 0.6394049525260925
Epoch 31, Loss: 0.6303167343139648
Epoch 32, Loss: 0.6216385960578918
Epoch 33, Loss: 0.6133355498313904
Epoch 34, Loss: 0.6053851842880249
Epoch 35, Loss: 0.5977634191513062
Epoch 36, Loss: 0.5904499292373657
Epoch 37, Loss: 0.5834182500839233
Epoch 38, Loss: 0.576655924320221
Epoch 39, Loss: 0.5701445937156677
Epoch 40, Loss: 0.5638659596443176
Epoch 41, Loss: 0.5578069686889648
Epoch 42, Loss: 0.551962673664093
Epoch 43, Loss: 0.5463167428970337
Epoch 44, Loss: 0.5408563613891602
Epoch 45, Loss: 0.5355707406997681
Epoch 46, Loss: 0.5304481387138367
Epoch 47, Loss: 0.5254788398742676
Epoch 48, Loss: 0.5206590294837952
Epoch 49, Loss: 0.5159763097763062
Epoch 50, Loss: 0.5114233493804932
Epoch 51, Loss: 0.5069987773895264
Epoch 52, Loss: 0.5026958584785461
Epoch 53, Loss: 0.49851468205451965
Epoch 54, Loss: 0.49444347620010376
Epoch 55, Loss: 0.49047645926475525
Epoch 56, Loss: 0.4866092801094055
Epoch 57, Loss: 0.4828360378742218
Epoch 58, Loss: 0.4791589081287384
Epoch 59, Loss: 0.4755704998970032
```

```
Epoch 60, Loss: 0.47206923365592957
Epoch 61, Loss: 0.46864813566207886
Epoch 62, Loss: 0.46530792117118835
Epoch 63, Loss: 0.4620462656021118
Epoch 64, Loss: 0.45885637402534485
Epoch 65, Loss: 0.455735981464386
Epoch 66, Loss: 0.4526822566986084
Epoch 67, Loss: 0.44969311356544495
Epoch 68, Loss: 0.44676828384399414
Epoch 69, Loss: 0.4439024031162262
Epoch 70, Loss: 0.4410918056964874
Epoch 71, Loss: 0.43833619356155396
Epoch 72, Loss: 0.4356333911418915
Epoch 73, Loss: 0.4329841732978821
Epoch 74, Loss: 0.4303843080997467
Epoch 75, Loss: 0.42783138155937195
Epoch 76, Loss: 0.42532551288604736
Epoch 77, Loss: 0.42286279797554016
Epoch 78, Loss: 0.420441210269928
Epoch 79, Loss: 0.41806069016456604
Epoch 80, Loss: 0.4157191812992096
Epoch 81, Loss: 0.41341933608055115
Epoch 82, Loss: 0.4111587107181549
Epoch 83, Loss: 0.40893620252609253
Epoch 84, Loss: 0.40675169229507446
Epoch 85, Loss: 0.4046034812927246
Epoch 86, Loss: 0.4024905264377594
Epoch 87, Loss: 0.40041017532348633
Epoch 88, Loss: 0.3983619213104248
Epoch 89, Loss: 0.3963446319103241
Epoch 90, Loss: 0.39435696601867676
Epoch 91, Loss: 0.3923986852169037
Epoch 92, Loss: 0.390468567609787
Epoch 93, Loss: 0.38856783509254456
Epoch 94, Loss: 0.3866945505142212
Epoch 95, Loss: 0.38484805822372437
Epoch 96, Loss: 0.38302746415138245
Epoch 97, Loss: 0.3812319338321686
Epoch 98, Loss: 0.37946003675460815
Epoch 99, Loss: 0.3777121305465698
```

In [22]:

```python
# explain the purpose of this code block by adding comments
# this code block evaluates the performance of the trained model
with torch.no_grad():
    x_test = x_test.to(device)  # Move test data to the device
    y_test = y_test.to(device)  # Move test labels to the device

    y_test_pred = mnist(x_test)  # Get predictions from the trained model
    _, predicted = torch.max(y_test_pred, 1)  # Get the predicted class for each
    correct = (predicted == y_test).sum().item()  # Count the number of correct
    accuracy = correct / len(y_test)  # Calculate the test accuracy
    print(f'Test Accuracy: {accuracy * 100:.2f}%')  # Print the test accuracy
```

```
Test Accuracy: 89.05%
```

In [23]:

```python
# show several test cases here, print the input number x_test and prediction res
for i in range(5):
    print(f"Sample {i+1}:")
    print("Input image:")
```

```
    print(x_test[i].squeeze().numpy())
    print(f"Prediction: {predicted[i].item()}\n")
```

```
Sample 1:
Input image:
[  0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.  84. 185. 159. 151.  60.  36.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0. 222. 254. 254. 254. 254. 241. 198. 198.
 198. 198. 198. 198. 198. 198. 170.  52.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.  67. 114.  72. 114. 163. 227. 254. 225.
 254. 254. 254. 250. 229. 254. 254. 140.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.  17.  66.  14.
  67.  67.  67.  59.  21. 236. 254. 106.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.  83. 253. 209.  18.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.  22. 233. 255.  83.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0. 129. 254. 238.  44.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.  59. 249. 254.  62.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0. 133. 254. 187.   5.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   9. 205. 248.  58.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0. 126. 254. 182.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
  75. 251. 240.  57.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.  19.
 221. 254. 166.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   3. 203.
 254. 219.  35.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.  38. 254.
 254.  77.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.  31. 224. 254.
 115.   1.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0. 133. 254. 254.
  52.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.  61. 242. 254. 254.
  52.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0. 121. 254. 254. 219.
  40.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0. 121. 254. 207.  18.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.]
Prediction: 7
```

```
Sample 2:
Input image:
[  0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0. 116. 125. 171. 255.
 255. 150.  93.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0. 169. 253. 253. 253. 253.
 253. 253. 218.  30.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0. 169. 253. 253. 253. 213. 142.
 176. 253. 253. 122.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.  52. 250. 253. 210.  32.  12.   0.
   6. 206. 253. 140.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.  77. 251. 210.  25.   0.   0.   0.
 122. 248. 253.  65.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.  31.  18.   0.   0.   0.   0.
 209. 253. 253.  65.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0. 117.
 247. 253. 198.  10.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.  76. 247.
 253. 231.  63.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0. 128. 253.
 253. 144.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0. 176. 246. 253.
 159.  12.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.  25. 234. 253. 233.
  35.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0. 198. 253. 253. 141.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.  78. 248. 253. 189.  12.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.  19. 200. 253. 253. 141.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0. 134. 253. 253. 173.  12.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0. 248. 253. 253.  25.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0. 248. 253. 253.  43.  20.  20.
  20.  20.   5.   0.   5.  20.  20.  37. 150. 150. 150. 147.  10.   0.
   0.   0.   0.   0.   0.   0.   0.   0. 248. 253. 253. 253. 253. 253.
 253. 253. 168. 143. 166. 253. 253. 253. 253. 253. 253. 253. 123.   0.
   0.   0.   0.   0.   0.   0.   0.   0. 174. 253. 253. 253. 253. 253.
 253. 253. 253. 253. 253. 253. 249. 247. 247. 169. 117. 117.  57.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0. 118. 123. 123. 123. 166.
 253. 253. 253. 155. 123. 123.  41.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.]
Prediction: 5
```

Sample 3:
Input image:

```
[  0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.  38. 254. 109.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.  87. 252.  82.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0. 135. 241.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.  45. 244. 150.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.  84. 254.  63.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0. 202. 223.  11.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
  32. 254. 216.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
  95. 254. 195.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
 140. 254.  77.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.  57.
 237. 205.   8.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0. 124.
 255. 165.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0. 171.
 254.  81.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.  24. 232.
 215.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0. 120. 254.
 159.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0. 151. 254.
 142.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0. 228. 254.
  66.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.  61. 251. 254.
  66.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0. 141. 254. 205.
   3.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.  10. 215. 254. 121.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   5. 198. 176.  10.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.]
```

Prediction: 1

Sample 4:
Input image:

```
[  0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.  11. 150.
 253. 202.  31.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.  37. 251.
 251. 253. 107.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.  21. 197. 251.
 251. 253. 107.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0. 110. 190. 251. 251.
 251. 253. 169. 109.  62.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0. 253. 251. 251. 251.
 251. 253. 251. 251. 220.  51.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0. 182. 255. 253. 253. 253.
 253. 234. 222. 253. 253. 253.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.  63. 221. 253. 251. 251. 251.
 147.  77.  62. 128. 251. 251. 105.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.  32. 231. 251. 253. 251. 220. 137.
  10.   0.   0.  31. 230. 251. 243. 113.   5.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.  37. 251. 251. 253. 188.  20.   0.
   0.   0.   0.   0. 109. 251. 253. 251.  35.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.  37. 251. 251. 201.  30.   0.   0.
   0.   0.   0.   0.  31. 200. 253. 251.  35.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.  37. 253. 253.   0.   0.   0.   0.
   0.   0.   0.   0.  32. 202. 255. 253. 164.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0. 140. 251. 251.   0.   0.   0.   0.
   0.   0.   0.   0. 109. 251. 253. 251.  35.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0. 217. 251. 251.   0.   0.   0.   0.
   0.   0.  21.  63. 231. 251. 253. 230.  30.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0. 217. 251. 251.   0.   0.   0.   0.
   0.   0. 144. 251. 251. 251. 221.  61.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0. 217. 251. 251.   0.   0.   0.   0.
   0. 182. 221. 251. 251. 251. 180.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0. 218. 253. 253.  73.  73. 228. 253.
 253. 255. 253. 253. 253. 253.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0. 113. 251. 251. 253. 251. 251. 251.
 251. 253. 251. 251. 251. 147.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.  31. 230. 251. 253. 251. 251. 251.
 251. 253. 230. 189.  35.  10.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.  62. 142. 253. 251. 251. 251.
 251. 253. 107.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.  72. 174. 251. 173.
  71.  72.  30.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.]
Prediction: 0
```

Sample 5:
Input image:

```
[  0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.  50. 224.   0.   0.
   0.   0.   0.   0.   0.  70.  29.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0. 121. 231.   0.   0.
   0.   0.   0.   0.   0. 148. 168.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   4. 195. 231.   0.   0.
   0.   0.   0.   0.   0.  96. 210.  11.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.  69. 252. 134.   0.   0.
   0.   0.   0.   0.   0. 114. 252.  21.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.  45. 236. 217.  12.   0.   0.
   0.   0.   0.   0.   0. 192. 252.  21.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0. 168. 247.  53.   0.   0.   0.
   0.   0.   0.   0.  18. 255. 253.  21.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.  84. 242. 211.   0.   0.   0.   0.
   0.   0.   0.   0. 141. 253. 189.   5.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0. 169. 252. 106.   0.   0.   0.   0.
   0.   0.   0.  32. 232. 250.  66.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.  15. 225. 252.   0.   0.   0.   0.   0.
   0.   0.   0. 134. 252. 211.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.  22. 252. 164.   0.   0.   0.   0.   0.
   0.   0.   0. 169. 252. 167.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   9. 204. 209.  18.   0.   0.   0.   0.
   0.   0.  22. 253. 253. 107.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0. 169. 252. 199.  85.  85.  85.  85.
 129. 164. 195. 252. 252. 106.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.  41. 170. 245. 252. 252. 252. 252.
 232. 231. 251. 252. 252.   9.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.  49.  84.  84.  84.  84.
   0.   0. 161. 252. 252.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0. 127. 252. 252.  45.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0. 128. 253. 253.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0. 127. 252. 252.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0. 135. 252. 244.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0. 232. 236. 111.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0. 179.  66.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.
   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.   0.]
```

Prediction: 4

In [ ]: